

Data Cleaning and Visualization using Titanic Dataset

Objective

The objective of this project is to clean a raw dataset and generate meaningful visual insights using Python.

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

Data Cleaning Steps

1. Loaded the Titanic dataset.
2. Identified missing values.
3. Filled missing values in the Age column using the median.
4. Filled missing values in the Embarked column using the mode.
5. Removed the Cabin column due to excessive missing values.
6. Removed duplicate records.

Visualizations Created

1. Survival Count
2. Survival by Gender
3. Age Distribution
4. Passenger Class vs Survival

Key Insights

- Female passengers had a higher survival rate than male passengers.
- First-class passengers had better survival rates.
- Most passengers were between 20 and 40 years old.
- Data preprocessing improved the quality of the dataset.

Conclusion

The project successfully demonstrated data cleaning, preprocessing, and visualization techniques using Python libraries.